

Measuring How Prompt Compression Changes LLM Network Traffic

A preliminary local-model experiment using LoCoMo workloads

	144	24	2
	valid packet captures	workload items	encrypted transports
Target model	Qwen3.5-9B served locally through vLLM		
Compression	Original prompt, LongLLMLingua 2x, LongLLMLingua 4x		
Workloads	16 long-context QA tasks and 8 speaker-event summary tasks		
Transports	TLS 1.3 / HTTP/1.1 over TCP and HTTP/3 over QUIC		
Scope	Controlled zero-RTT loopback baseline, one capture per cell		

Main conclusion

Prompt compression substantially reduces network traffic when the prompt is the dominant payload. Its end-to-end benefit is smaller when the generated response dominates the exchange.

1 Executive summary

The completed pilot establishes an end-to-end measurement pipeline that can compress long conversational prompts, send them to a local LLM over two encrypted application transports, capture the traffic, and recover comparable per-request measurements. All **144 planned baseline captures are valid**.

Finding	Measured result	Interpretation
QA traffic	71–73% lower total bytes at 4x	The long prompt dominates the exchange, so compression directly removes most transmitted content.
Summary traffic	20–22% lower total bytes at 4x	Upload shrinks, but a long generated summary remains the largest payload.
Token-to-byte relation	Pearson $r = 0.88$ for TLS and HTTP/3	Input token count is a strong predictor of client-to-server traffic.
Protocol comparison	Total bytes stay within 4% of parity	On loopback, protocol choice changes packetization more than total bytes or latency.

1.1 What has been completed

- Prepared LoCoMo-derived QA and speaker-event summarization workloads.
- Produced deterministic original, 2x, and 4x prompt variants.
- Served Qwen3.5-9B locally with vLLM on two approximately 32 GB GPUs.
- Implemented TLS 1.3 and HTTP/3 test paths and packet capture.
- Added finish-reason and token accounting to distinguish complete from truncated generations.
- Executed and recovered all baseline jobs, including previously failed jobs, while preserving resumability.
- Merged the four traffic metric tables into one 144-row artifact.
- Added a deterministic analysis pipeline with paired ratios, bootstrap confidence intervals, and conference-style figures.

2 Experimental design and settings



Setting	Value used in the baseline experiment
Target LLM	Qwen3.5-9B, served by vLLM
Hardware	Two GPU workers; each worker handled a disjoint sample shard
Decoding	Temperature 0, seed 42, thinking disabled
QA generation cap	256 output tokens
Summary generation cap	1,024 output tokens
Prompt conditions	No compression, LongLLMLingua 2x, LongLLMLingua 4x
TLS path	TLS 1.3 with HTTP/1.1 over TCP
QUIC path	HTTP/3 over QUIC
Capture unit	One request-response transaction per metric record
Network condition	Local loopback baseline with effectively zero added RTT
Repetitions	One completed repetition for each baseline cell
Primary metrics	Directional bytes, total bytes, packet count, elapsed time, input/output tokens, and finish reason

2.1 Run-count logic

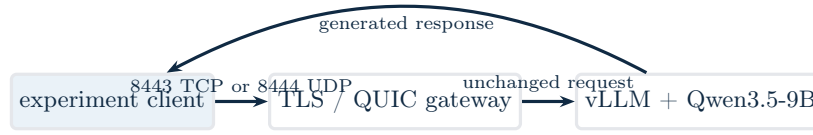
$$(16 \text{ QA} + 8 \text{ summary}) \times 3 \text{ prompt forms} \times 2 \text{ transports} \times 1 \text{ repeat} = \mathbf{144 \text{ captures}}$$

The planned full controlled sweep expands to 52 workload items, two network conditions, and three repetitions:

$$52 \times 3 \times 2 \times 2 \times 3 = \mathbf{1,872 \text{ runs.}}$$

2.2 How traffic is captured

The local client sends the same OpenAI-compatible request payload through two front-door listeners. Caddy terminates TLS 1.3 and forwards the unchanged request to vLLM. TCP port 8443 carries HTTP/1.1; UDP port 8444 carries HTTP/3-only QUIC traffic through an aioquic client. There is no TCP fallback in the HTTP/3 condition.



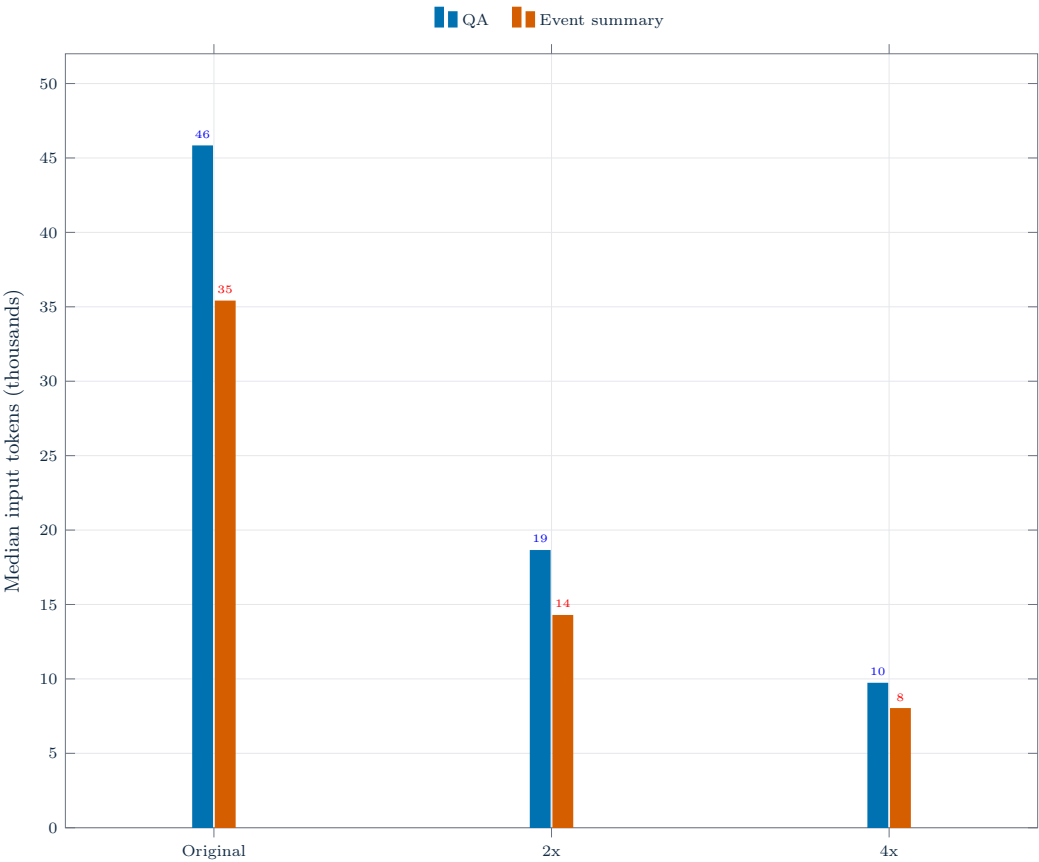
`dumpcap` performs lightweight acquisition and writes one `pcapng` file per request. `tshark` then calculates directional bytes, total frame bytes, packet counts, and protocol evidence offline. This separates acquisition from analysis and permits metrics to be regenerated without repeating GPU inference.

2.3 How model randomness and multi-turn dependence are controlled

- Temperature 0 and seed 42 reduce decoding variability.
- Every condition receives the same frozen conversation and task instruction; only the prompt representation changes.
- QA tasks do not create a new branching conversation. Each question is evaluated independently against the original recorded LoCoMo history.
- Event summarization similarly requests a fixed speaker-level artifact from the recorded history. It does not feed a generated answer into the next sample.
- The measured response is retained with its token count and finish reason, so unexpectedly short or truncated outputs can be identified.

3 Compression reaches distinct token budgets

LongLLMLingua produced clearly separated prompt sizes for both tasks. This confirms that the requested conditions are operationally distinct before any traffic comparison is made.

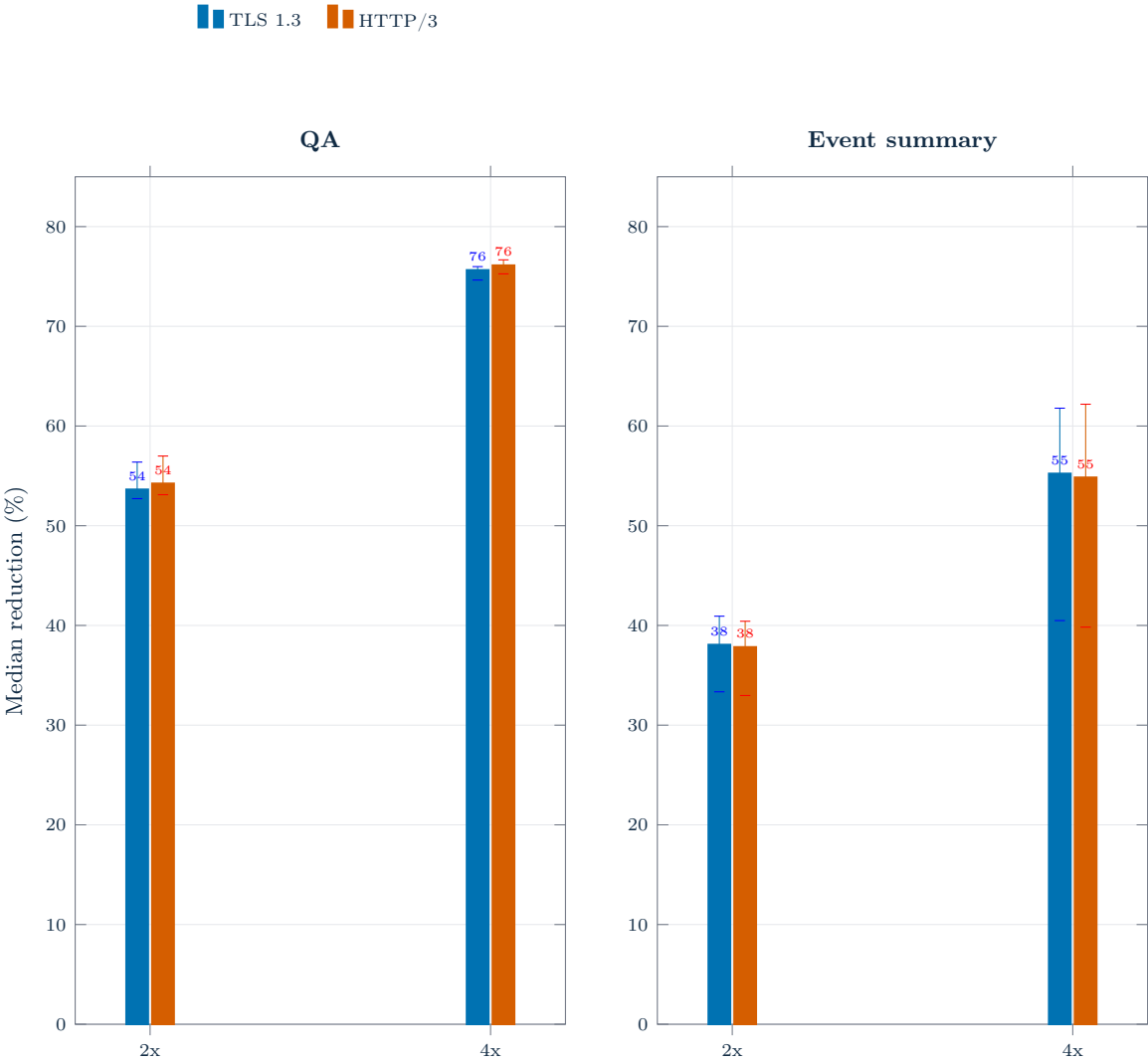


Workload	Original	2x	4x
QA median input tokens	45,752	18,568	9,650
Event-summary median input tokens	35,326	14,204	7,950

Table 1. Median input-token counts. Transport is omitted because the same prompt is used for both transport paths.

4 Upload traffic falls after prompt compression

Client-to-server traffic follows the reduced prompt size closely. The result is consistent across TLS 1.3 and HTTP/3.

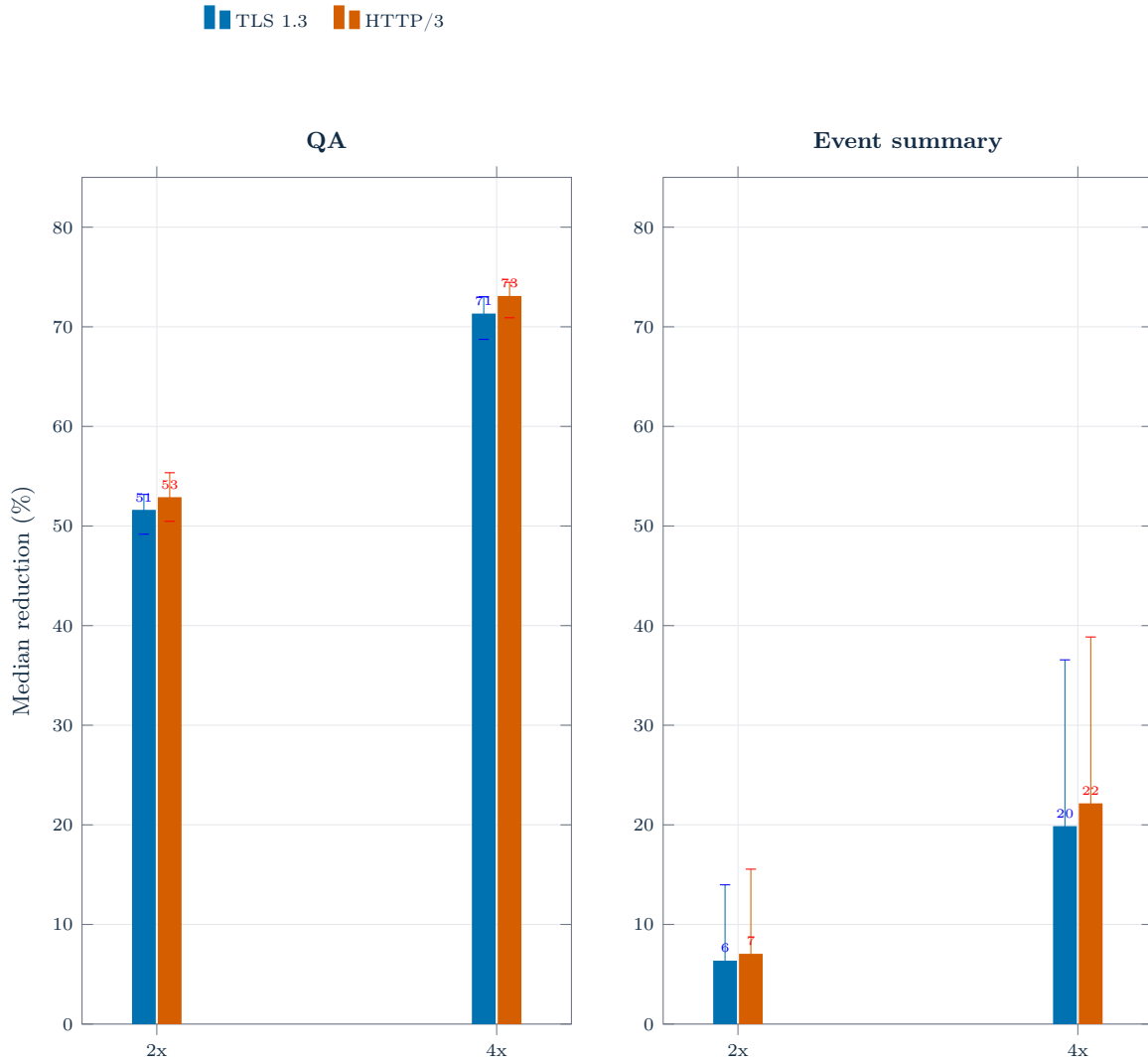


Workload	Transport	2x reduction	4x reduction
QA	TLS 1.3	53.6%	75.6%
QA	HTTP/3	54.2%	76.1%
Event summary	TLS 1.3	38.0%	55.2%
Event summary	HTTP/3	37.8%	54.8%

Table 2. Paired median reduction in client-to-server bytes versus the original prompt. Whiskers in the figure are bootstrap 95% confidence intervals.

5 Total traffic reduction depends on workload direction

For QA, upload dominates and the traffic benefit survives end-to-end. For event summarization, long generated responses dilute the benefit of reducing the prompt.

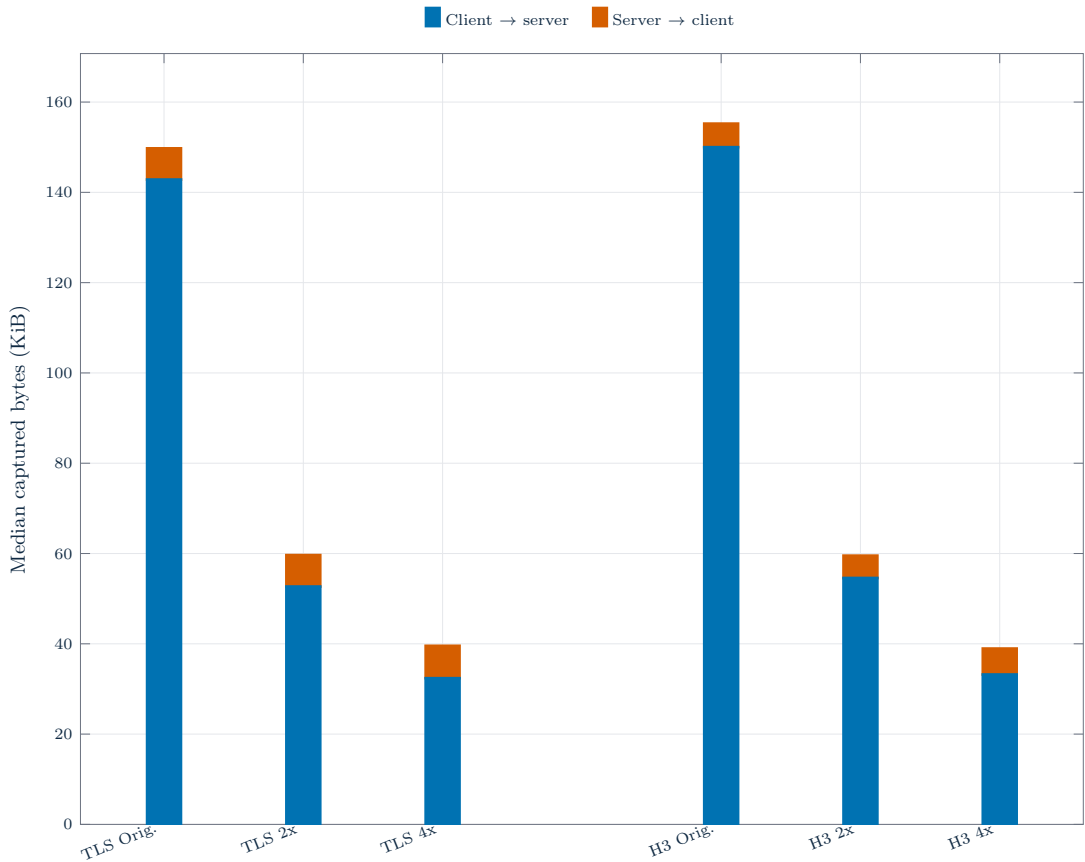


Workload	Transport	2x reduction	4x reduction
QA	TLS 1.3	51.5%	71.2%
QA	HTTP/3	52.7%	72.9%
Event summary	TLS 1.3	6.2%	19.7%
Event summary	HTTP/3	6.9%	22.0%

Table 3. Paired median reduction in bidirectional frame bytes.

6 QA is an input-dominated exchange

The uploaded conversation is the largest component of QA traffic. Output answers are short, so prompt compression removes most of the transmitted payload.

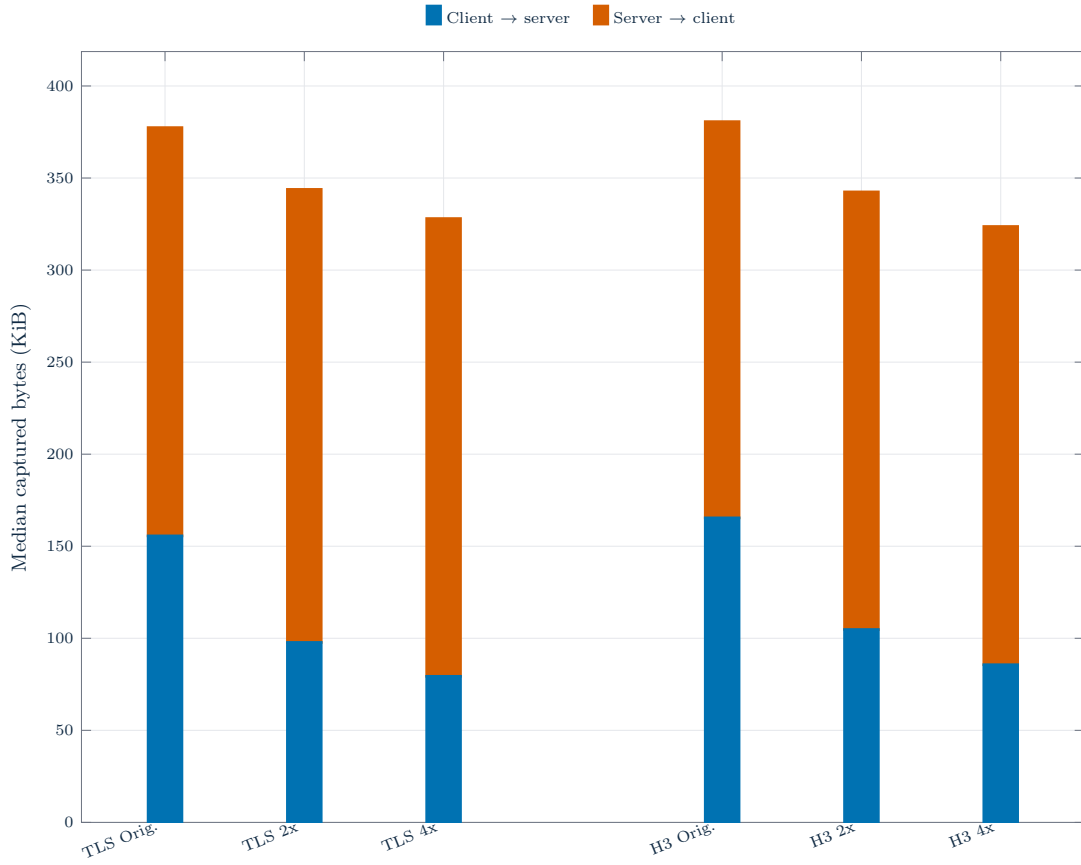


Transport	Prompt	Upload (KiB)	Download (KiB)	Total (KiB)
TLS 1.3	Original	142.8	6.9	147.9
TLS 1.3	2x	52.7	7.0	62.2
TLS 1.3	4x	32.4	7.2	41.1
HTTP/3	Original	150.0	5.2	154.2
HTTP/3	2x	54.6	4.9	62.3
HTTP/3	4x	33.2	5.7	39.8

Table 4. Median directional and total captured bytes for QA.

7 Event summaries are output-dominated exchanges

Compression still reduces upload traffic, but the generated summary remains roughly 220–255 KiB. Slightly longer compressed-prompt outputs further limit total-byte savings.

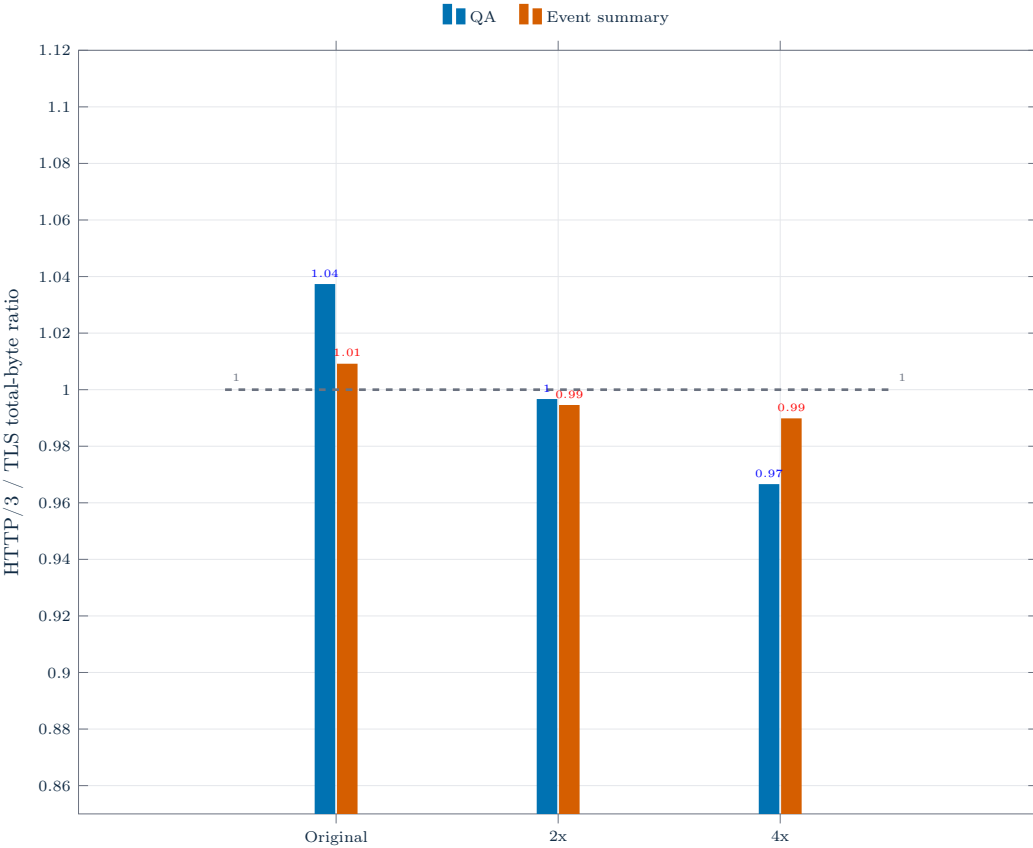


Transport	Prompt	Upload (KiB)	Download (KiB)	Total (KiB)
TLS 1.3	Original	155.6	221.8	378.2
TLS 1.3	2x	97.7	246.1	342.6
TLS 1.3	4x	79.3	248.6	330.5
HTTP/3	Original	165.4	215.2	381.7
HTTP/3	2x	104.8	237.7	342.5
HTTP/3	4x	85.6	238.0	326.1

Table 5. Median directional and total captured bytes for event summaries.

8 TLS and HTTP/3 transfer similar total bytes

On the zero-RTT loopback baseline, all median HTTP/3-to-TLS total-byte ratios remain within four percent of parity.

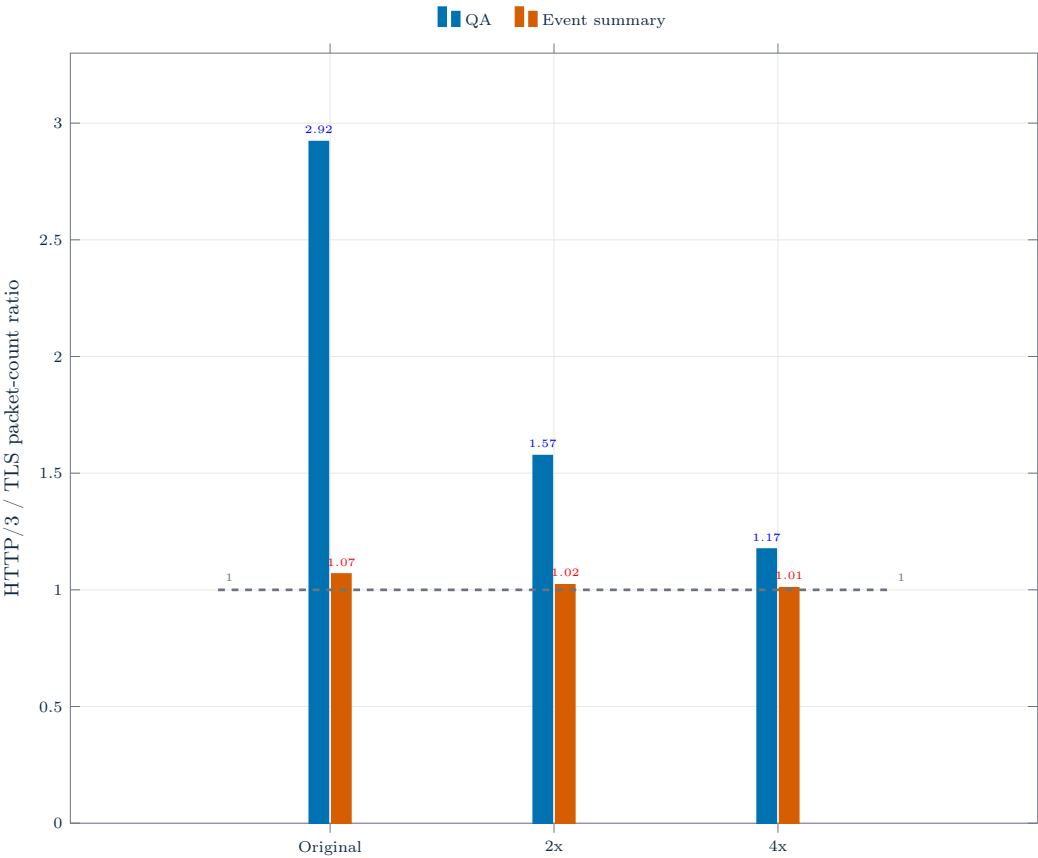


Workload	Original	2x	4x
QA: HTTP/3 / TLS total bytes	1.04	1.00	0.97
Summary: HTTP/3 / TLS total bytes	1.01	0.99	0.99

Table 6. Paired median protocol ratios. A ratio of 1.00 indicates parity.

9 HTTP/3 packetization is visible for short QA

HTTP/3 sends more captured packets for the short QA exchange, especially for the original long prompt. This difference narrows as compression reduces the payload. Summary packet counts are close between transports.

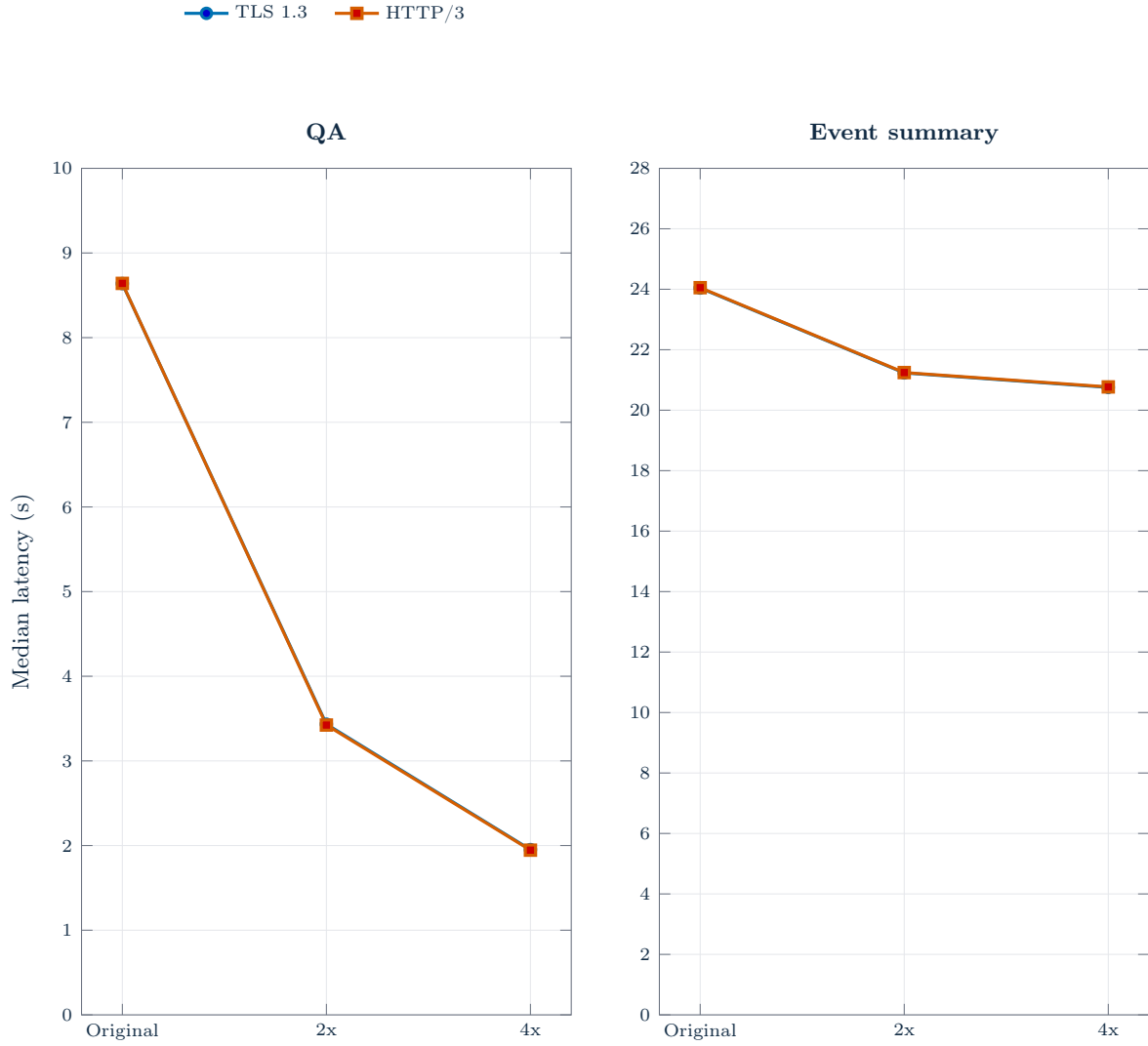


Workload	Original	2x	4x
QA: HTTP/3 / TLS packets	2.92	1.57	1.17
Summary: HTTP/3 / TLS packets	1.07	1.02	1.01

Table 7. Paired median packet-count ratios.

10 Latency follows the amount of input processed

QA latency falls strongly because the target model processes far fewer input tokens. Summary latency falls less because response generation remains the dominant computation.



Workload	Transport	Original	2x	4x
QA	TLS 1.3	8.64 s	3.43 s	1.95 s
QA	HTTP/3	8.64 s	3.42 s	1.95 s
Event summary	TLS 1.3	24.04 s	21.23 s	20.75 s
Event summary	HTTP/3	24.05 s	21.24 s	20.77 s

Table 8. Median end-to-end generation latency.

11 Interpretation, research standard, and next actions

11.1 What the pilot establishes

1. **The measurement chain works end-to-end.** Compression, local inference, encrypted transmission, packet capture, metric extraction, retry, and resume all produced a complete baseline matrix.
2. **Network savings are workload-dependent.** A compression ratio alone does not determine total savings; the relative size of request and response traffic matters.
3. **The transport baseline is controlled.** TLS and HTTP/3 receive the same semantic input and model configuration, enabling paired comparison.
4. **The current results are preliminary, not a final network-condition claim.** They represent one loopback repetition and should be treated as an effect-discovery and pipeline-validation study.

11.2 Recommended full experiment

Factor	Recommended setting
Workloads	52 total workload items, retaining both QA and event summarization
Compression	Original, LongLLMLingua 2x, LongLLMLingua 4x
Transports	TLS 1.3 / HTTP/1.1 and HTTP/3 / QUIC
Network conditions	Controlled loopback plus one realistic RTT condition
RTT treatment	Fixed delay applied symmetrically and reported explicitly
MTU	Fixed value matched across transports and documented with the result
Offloads	Disable segmentation and aggregation offloads during capture
Connection state	Report whether measurements use warm persistent connections or cold per-request connections; do not mix them in one estimate
Repetitions	Three independent repetitions per experiment cell
Primary analysis	Paired medians, dispersion or confidence intervals, and per-sample ratios

11.3 Why three repetitions are defensible

Three repetitions are a practical minimum for a costly GPU and packet-capture experiment. They permit variability checks and paired aggregation without the cost of ten repetitions. For a top-tier submission, the report should show confidence intervals, disclose that repetitions are limited, and emphasize effect sizes rather than relying only on significance tests. Sample diversity across 52 workload items contributes more evidence than repeatedly measuring a small number of prompts ten times.

11.4 Resume and partial-result strategy

The runner should treat each combination of sample, compression condition, transport, network condition, and repetition as a unique job key. A completed job is appended atomically to the result artifact. On restart, existing valid keys are skipped, failed keys are re-queued, and new results are merged without overwriting the 144 baseline records. This permits a two-hour partial run for a presentation while preserving all work for later continuation.

12 Figure-selection guide

The separate figure catalog contains thirteen candidate views. The following subset is recommended for the main paper or presentation.

Figure	Purpose	Recommendation
F1	Input-token medians	Keep as a setup validation figure or appendix item.
F2	Upload-byte reduction	Keep; this directly answers the traffic question.
F3	Total-byte reduction	Keep; this is the clearest primary result.
F4–F5	Directional stacked bytes	Keep both or combine as a two-panel figure; they explain the workload-dependent effect.
F6	HTTP/3-to-TLS byte ratio	Keep for the protocol comparison.
F7	HTTP/3-to-TLS packet ratio	Keep if packetization is a research question; otherwise move to the appendix.
F8	Absolute latency	Keep if system performance is discussed.
F9	Token/upload scatter	Strong appendix evidence for the $r = 0.88$ relation.
F10	Paired ECDF	Best paper-style distribution figure; keep in the paper.
F11–F13	Output share, absolute traffic, latency reduction	Useful alternate views; choose only one to avoid repeating the same conclusion.

13 References and reproducibility

- Pan et al. “LLMLingua-2: Data Distillation for Efficient and Faithful Task-Agnostic Prompt Compression.” arXiv:2403.12968, 2024.
- Maharana et al. “Evaluating Very Long-Term Conversational Memory of LLM Agents.” Proceedings of ACL 2024.
- Metric artifact: `artifacts/preliminary_metrics_baseline.csv`.
- Analysis: `analysis/prepare_figure_data.py`.
- Figure definitions: `figures/conference_plots.tex`.

All numerical claims in this report are computed from the 144 valid baseline captures. Tables use medians unless explicitly described as paired median ratios. Bootstrap confidence intervals use a fixed random seed for reproducibility.